

List Programs

1. Find the largest element in a list

Write a program that takes a list of numbers and prints the **largest number**.

Example

Input: [4, 7, 2, 9, 5]

Output: 9

2. Find the sum of elements in a list

Write a program to **calculate the sum of all elements** in a list.

3. Remove duplicates from a list

Write a program to **remove duplicate elements** from a list.

Example

Input: [1, 2, 2, 3, 4, 4, 5]

Output: [1, 2, 3, 4, 5]

4. Reverse a list

Write a program to **reverse the elements of a list**.

Example

Input: [1, 2, 3, 4]

Output: [4, 3, 2, 1]

5. Find even numbers in a list

Write a program that **prints all even numbers** from a list.

Tuple Programs

6. Find the length of a tuple

Write a program that **takes a tuple and prints its length.**

7. Convert tuple to list

Write a program to **convert a tuple into a list.**

8. Count occurrences in a tuple

Write a program to **count how many times a number appears in a tuple.**

Example

Tuple: (1, 2, 3, 2, 4, 2)

Find count of 2.

9. Find maximum element in a tuple

Write a program to **find the largest element in a tuple.**

10. Convert list to tuple

Write a program that **converts a list into a tuple.**

Set Programs

11. Find union of two sets

Write a program to **find the union of two sets.**

Example

A = {1, 2, 3}

B = {3, 4, 5}

Output: {1, 2, 3, 4, 5}

12. Find intersection of two sets

Write a program to **find common elements between two sets**.

13. Remove element from a set

Write a program that **removes an element from a set**.

14. Check if element exists in set

Write a program to **check whether a number exists in a set**.

15. Find difference of two sets

Write a program to **find elements present in set A but not in set B**.

String Programs

16. Reverse a string

Write a program to **reverse a string**.

Example

Input: "Python"

Output: "nohtyP"

17. Count vowels in a string

Write a program that **counts vowels (a, e, i, o, u)** in a string.

18. Check palindrome string

Write a program to **check if a string is a palindrome**.

Example

Input: "madam"

Output: Palindrome

19. Count characters in a string

Write a program to **count the number of characters in a string**.

20. Convert string to uppercase

Write a program to **convert a string into uppercase**.

Dictionary Programs

21. Create a dictionary and print keys and values

Write a program that **creates a dictionary and prints all keys and values**.

Example

```
student = {"name": "John", "age": 20, "course": "CS"}
```

22. Add a new key-value pair

Write a program to **add a new item to a dictionary**.

23. Check if a key exists in dictionary

Write a program to **check if a key exists** in a dictionary.

24. Find sum of dictionary values

Write a program that **calculates the sum of all values** in a dictionary.

Example

```
d = {"a":10, "b":20, "c":30}
```

Output: 60

25. Merge two dictionaries

Write a program to **merge two dictionaries into one**.

Mixed Practice Questions

26. Remove duplicates from a list using a set.

27. Count frequency of characters in a string using dictionary.

Example

Input: "banana"

Output

```
b:1  
a:3  
n:2
```

28. Find common elements between two lists using set.

29. Convert a string into a list of words.

Example

Input: "Python is easy"

Output

```
['Python', 'is', 'easy']
```

30. Store student marks in dictionary and find average marks.

Example

```
marks = {"Math":80,"Science":90,"English":70}
```

Output: Average marks.

31. Write a program that takes two sets from the user and prints their symmetric difference.

Example

Input

```
Set1: 1 2 3 4  
Set2: 3 4 5 6
```

Output

```
{1, 2, 5, 6}
```

32. Write a program to check if two sets have no common elements using symmetric difference.

33. Write a Python program to find the intersection of two sets using `intersection_update()`.

Example

```
A = {1, 2, 3, 4}  
B = {3, 4, 5, 6}
```

34. Write a program that removes all elements from the first set that are not present in the second set using `intersection_update()`.

35. Write a program that takes two sets from the user and performs all set operations:

- Union
- Intersection
- Difference
- Symmetric Difference

36. Write a program to print the sum and average of first n natural numbers.

37. Write a program to print multiplication table by getting the value from user

38. Write a Program to extract each digit from an integer in the reverse order.

39. Write a Python program to get the difference between a given number and 17, if the number is greater than 17 return double the difference.

40. Write a Python program to compute the greatest common divisor (GCD) of two positive integers.

41. Write a program that will perform the following:

- a) Read the marks of three subjects: Computer Science, Mathematics and Physics, out of 100
- b) Calculate the aggregate marks
- c) Calculate the percentage of marks

42. Write a program to print the square of first n natural numbers.

43. Write a program to check whether a given number is an Armstrong number. An Armstrong number of three digits is an integer such that the sum of the cubes of its digits is equal to the number itself. For example, 371 is an Armstrong number since $3^3 + 7^3 + 1^3 = 371$

44. A school has following rules for grading system:

- a. Below 25 - F
- b. 25 to 45 - E

- c. 45 to 50 - D
- d. 50 to 60 - C
- e. 60 to 80 - B
- f. Above 80 - A

Ask user to enter marks and print the corresponding grade.

45. A shop will give discount of 10% if the cost of purchased quantity is more than 1000.

Ask user for quantity

Suppose, one unit will cost 100.

Print total cost for user.

46. A student will not be allowed to sit in exam if his/her attendance is less than 75%.

Take following input from user

Number of classes held

Number of classes attended.

And a) print percentage of class attended. b) Is student allowed to sit in exam or not. Print allowed or not allowed